

C Programming - Menu Driven Array Operations

Batch - 15

Name: Dhruv Pandey | SAP ID: 590037660

Problem Statement

Write a C program to perform the following menu-driven operations:

1	Count even numbers, odd numbers, positive, negative, and zero elements in an array.
2	Search for an element in an array using linear search.
3	Merge two arrays.
4	Exit.

Algorithm

1. Start the program and display the menu.
2. Read the user's choice.
3. If choice is 1, read an array and count even, odd, positive, negative, and zero elements.
4. If choice is 2, read an array and the search key, then use linear search to find the element.
5. If choice is 3, read two arrays, copy their elements into a third array, and display the merged array.
6. If choice is 4, terminate the program.
7. For any other choice, display an invalid-choice message.
8. Repeat the menu until the user selects Exit.

C Program

```
#include <stdio.h>

void countElements()
{
    int a[100], n, i;
    int even = 0, odd = 0, positive = 0, negative = 0, zero = 0;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter %d elements:\n", n);
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    for (i = 0; i < n; i++)
    {
        if (a[i] % 2 == 0)
            even++;
        else
            odd++;

        if (a[i] > 0)
            positive++;
        else if (a[i] < 0)
            negative++;
        else
            zero++;
    }

    printf("Even elements = %d\n", even);
}
```

```

    printf("Odd elements = %d\n", odd);
    printf("Positive elements = %d\n", positive);
    printf("Negative elements = %d\n", negative);
    printf("Zero elements = %d\n", zero);
}

void linearSearch()
{
    int a[100], n, i, key, found = 0;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter %d elements:\n", n);
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    printf("Enter element to search: ");
    scanf("%d", &key);

    for (i = 0; i < n; i++)
    {
        if (a[i] == key)
        {
            printf("%d found at position %d\n", key, i + 1);
            found = 1;
            break;
        }
    }

    if (found == 0)
        printf("%d not found in the array.\n", key);
}

void mergeArrays()
{
    int a[100], b[100], c[200];
    int n1, n2, i;

    printf("Enter number of elements in first array: ");
    scanf("%d", &n1);

    printf("Enter %d elements of first array:\n", n1);
    for (i = 0; i < n1; i++)
        scanf("%d", &a[i]);

    printf("Enter number of elements in second array: ");
    scanf("%d", &n2);

    printf("Enter %d elements of second array:\n", n2);
    for (i = 0; i < n2; i++)
        scanf("%d", &b[i]);

    for (i = 0; i < n1; i++)
        c[i] = a[i];

    for (i = 0; i < n2; i++)
        c[n1 + i] = b[i];

    printf("Merged array:\n");
    for (i = 0; i < n1 + n2; i++)
        printf("%d ", c[i]);

    printf("\n");
}

int main()
{
    int choice;

    do
    {
        printf("\n--- ARRAY OPERATIONS MENU ---\n");
        printf("1. Count even, odd, positive, negative and zero elements\n");
        printf("2. Linear search\n");
        printf("3. Merge two arrays\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch (choice)

```

```
{
    case 1:
        countElements();
        break;

    case 2:
        linearSearch();
        break;

    case 3:
        mergeArrays();
        break;

    case 4:
        printf("Exiting program...\n");
        break;

    default:
        printf("Invalid choice. Please enter 1 to 4.\n");
}
} while (choice != 4);

return 0;
}
```

Sample Test Cases

Option	Input / Example	Expected Result
1	Array: -2, 0, 3, 4, -5	Even = 3, Odd = 2, Positive = 2, Negative = 2, Zero = 1
2	Array: 10, 20, 30, 40; Key: 30	30 found at position 3
2	Array: 10, 20, 30; Key: 50	50 not found in the array
3	A = {1, 2, 3}; B = {4, 5}	Merged array: 1 2 3 4 5
4	Choice = 4	Program exits

Result

The menu-driven C program performs counting, linear search, array merging, and exit operations successfully using separate functions and a switch statement.

Execution (Sample Run)

The following terminal screenshot shows the program compiled and executed successfully.

```
Last login: Tue Sep 22 18:35:40 on console
dhruvpandey@DHRUVs-MacBook-Air ~ % nano day29.c
dhruvpandey@DHRUVs-MacBook-Air ~ % clang day29.c -oday29
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day29

--- ARRAY OPERATIONS MENU ---
1. Count even, odd, positive, negative and zero elements
2. Linear search
3. Merge two arrays
4. Exit
Enter your choice: 1
Enter number of elements: 5
Enter 5 elements:
-2 0 3 4 -5
Even elements = 3
Odd elements = 2
Positive elements = 2
Negative elements = 2
Zero elements = 1

--- ARRAY OPERATIONS MENU ---
1. Count even, odd, positive, negative and zero elements
2. Linear search
3. Merge two arrays
4. Exit
Enter your choice: █
```

Result

The menu-driven array program executed successfully for the displayed test case.